Dynamic Pricing AI

Autonomous E-Commerce Revenue Optimization

Combining Demand Forecasting & Reinforcement Learning

CME4416 - Introduction to Data Mining

GROUP 5

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Executive Summary

In today's hyper-competitive digital retail landscape, static pricing is obsolete. Our project presents an end-to-end autonomous dynamic pricing engine.
The system leverages **Deep Reinforcement Learning (DRL)** to adjust prices in real-time based on competitor actions, seasonality, and consumer price elasticity.
Key Innovation: A **"Model-in-the-Loop"** architecture where a Random Forest simulator serves as the training environment for a PPO agent.



Presentation Agenda

Talha

Data Engineering & Environment Simulation Phase 1: MySQL database architecture, synthetic data generation (`setup_db.py`), modeling price elasticity, seasonality, and competitor stochastic behaviors.

Durukan

RL Policy Optimization Phase 3: Custom Gymnasium environment design, reward function formulation (profit maximization), PPO (Stable-Baselines3) agent training and backtesting.

Ertuğrul

Phase 2: Exploratory data analysis (EDA), feature engineering, Random Forest Regressor training, model serialization, and baseline accuracy validation.

Emirhan

Full-Stack Integration & API Phase 4: FastAPI REST endpoints, SQLAlchemy ORM management, Next.js interactive dashboard development, Recharts integration for live market visualization.

Presentation Agenda

01. Introduction

Problem statement and the strategic need for dynamic pricing.

02. System Architecture

Technical breakdown of the 3-layer modular stack.

03. AI Methodology

Random Forest forecasting and RL policy optimization.

04. Experimental Results

Quantitative analysis of revenue and profit lift.

Section I: Introduction

The Economics of E-Commerce Pricing in the AI Era

E-Commerce Market Volatility

Key Market Drivers

- **Aggressive Competition:** Platforms like Amazon adjust prices millions of times daily.
- **Consumer Behavior:** Rapid shifts in demand due to trends and social proof.
- **Stochastic Externalities:** Unpredictable competitor stock levels and pricing wars.



The Problem: Static Pricing Failures

Revenue Loss

Setting prices too high results in inventory stagnation and lost market share.

Margin Erosion

Setting prices too low devalues the brand and leaves money on the table.

"Traditional rule-based systems are too rigid to capture non-linear market sensitivities."

Motivation: Why Autonomous AI?



Real-time Speed

Processing millions of scenarios to adjust prices in seconds.



Long-term Strategy

Optimizing for cumulative profit, not just immediate sales.



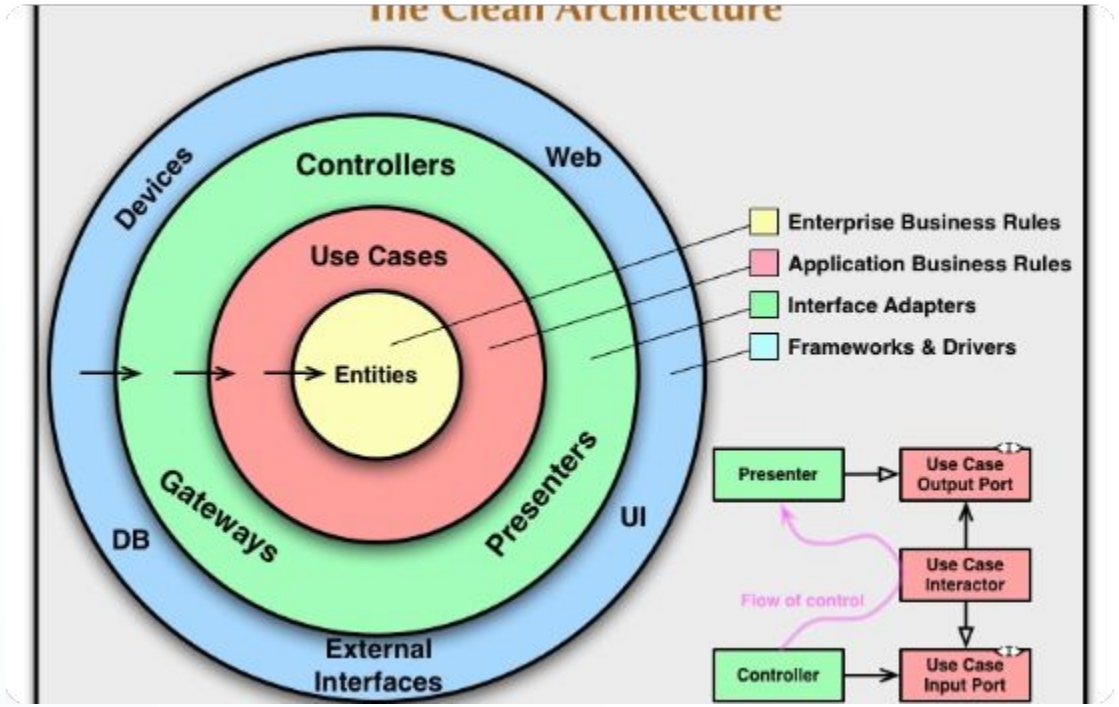
Adaptive Learning

Continuous refinement of the pricing policy as market data evolves.

Section II: System Architecture

Modular Full-Stack Infrastructure for Scalable Pricing

High-Level System Diagram



Unified Data Flow

Our system synchronizes high-fidelity database observations with real-time AI inference.

Database: MySQL engine for transactional persistence.

Brain: FastAPI serving optimized pricing policies.

Interface: Next.js dashboard for human-in-the-loop oversight.

Layer 1: The Data Engine

MySQL Infrastructure

73,000 Records: 2 years of daily sales for 100 SKUs.

Product Meta: Base cost, category, and historical current price.

Market Data: Real-time tracking of competitor pricing shifts.



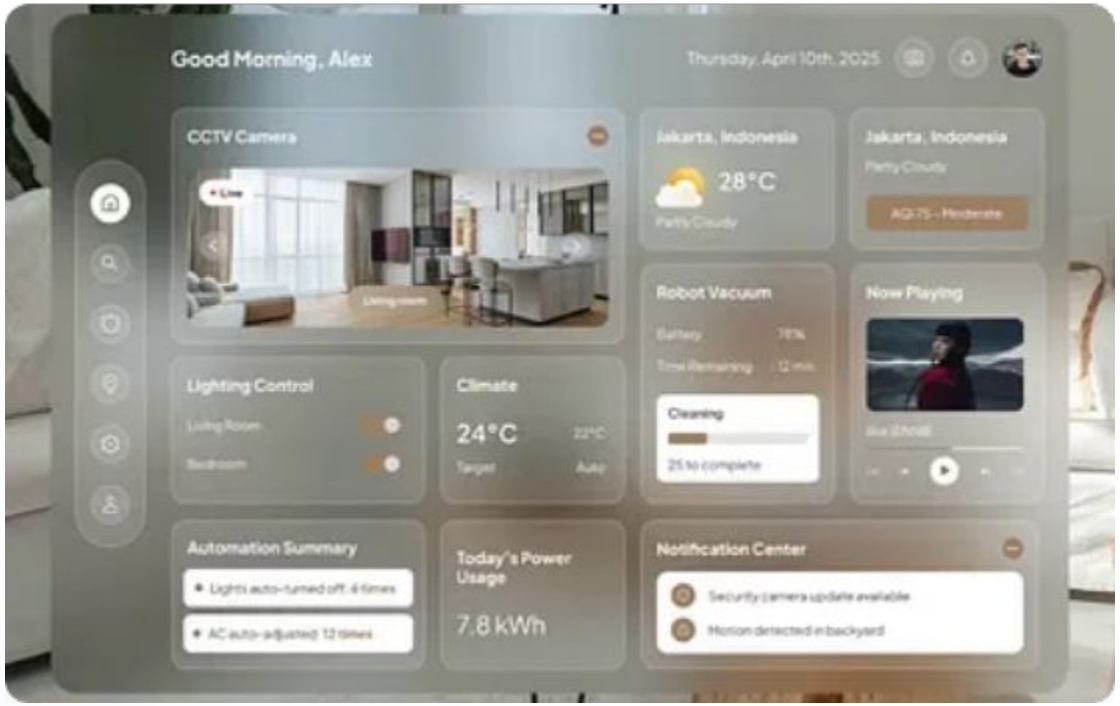
Layer 2: AI Backend Services

FastAPI & ML Framework

Our backend serves as the orchestration layer for the dual-model system.

- **Asynchronous Serving:** Handling pricing requests at scale.
- **Model Persistence:** Joblib-serialized Scikit-learn and Stable-Baselines3 weights.
- **Gymnasium Integration:** Hosting the virtual market environment.

Layer 3: Analytical Dashboard



Next.js + Tailwind Aesthetics

- **Global Overview:** Real-time revenue and SKU performance.
- **Optimizer View:** Manual triggers for AI price recommendations.
- **Dynamic Themes:** Light/Dark mode for presentation clarity.

Section III: Data Mining & Simulation

Building a High-Fidelity Synthetic Market Environment

High-Fidelity Synthetic Generation

Since real-world dynamic pricing datasets are proprietary, we mined economic principles to generate a 73,000-row synthetic dataset.

$$D_t = f_{elast}(P_t, P_{comp}) + \text{Seasonality} + \text{Noise}$$

This ensures the AI learns **causal relationships**, not just correlation.

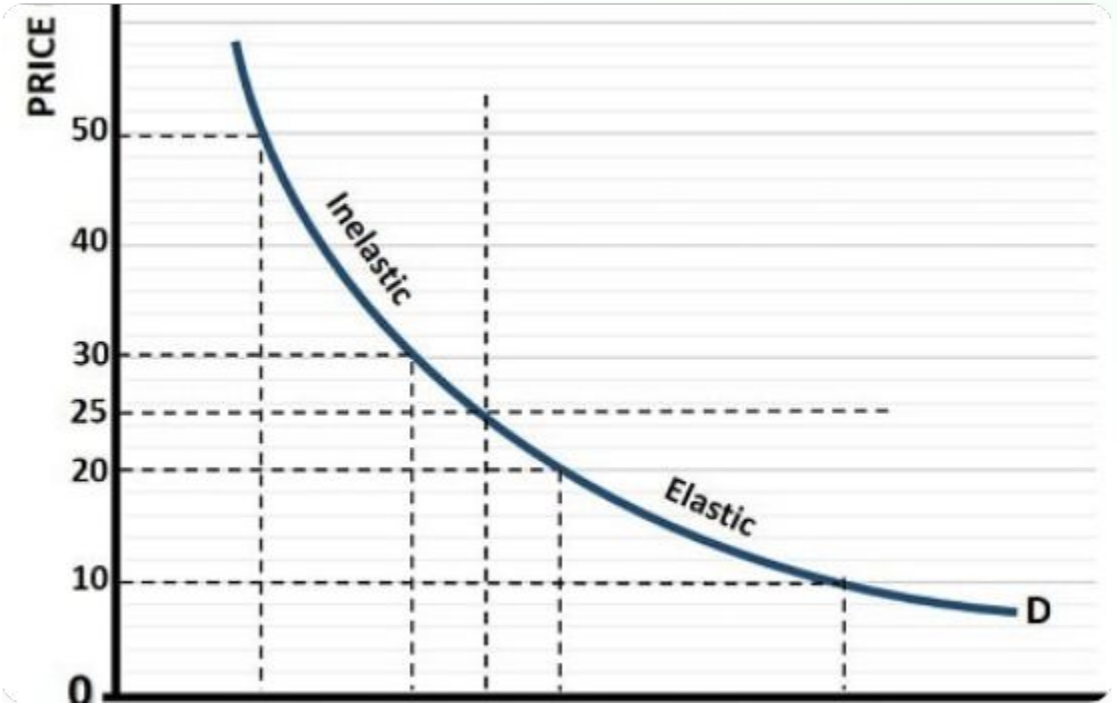
Modeling Price Elasticity

Consumer Sensitivity

Every SKU is assigned a unique coefficient representing how demand responds to price shifts.

- **Electronics:** High elasticity (Price sensitive).
- **Home Essentials:** Low elasticity (Necessity).

The system mines these patterns to find the "sweet spot" where revenue meets volume.



Stochastic Competitor Simulation

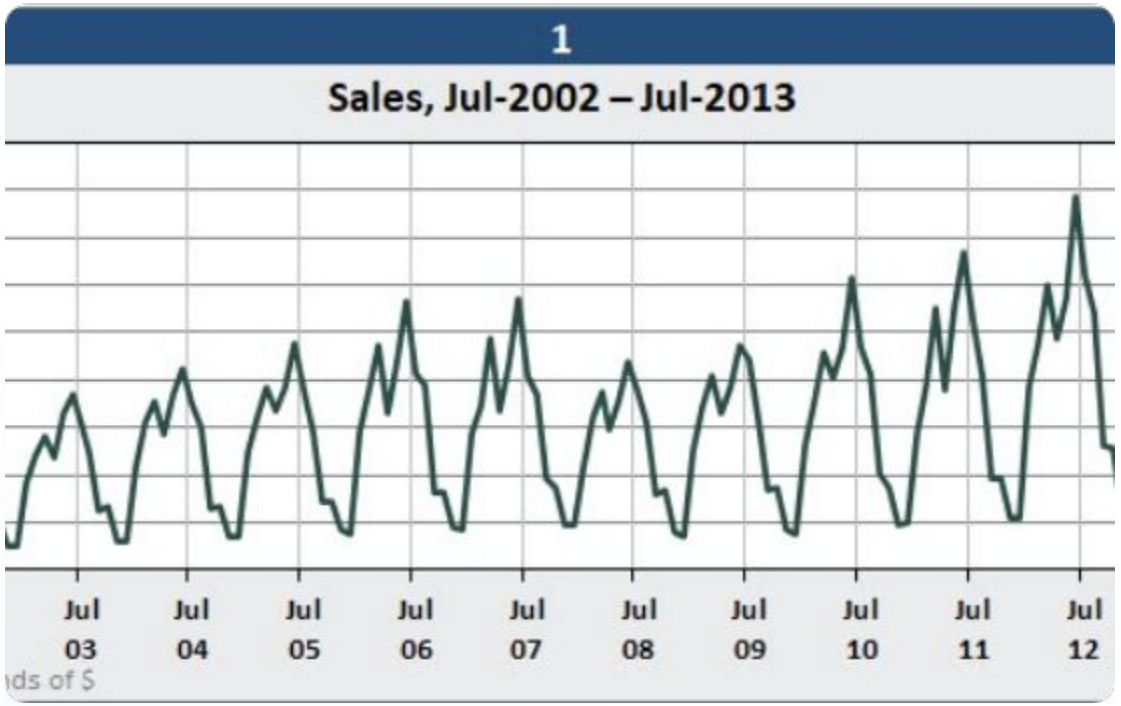
Random Walk Prices

Competitors adjust prices based on random fluctuations, forcing our AI to stay competitive without predictable patterns.

Trend Following

Competitors mirror seasonal market trends, increasing the complexity of the optimization state space.

Temporal Trend Mining



Dynamic Seasonality

Sales spikes are modeled to reflect real-world events:

- **Holiday Sales:** Massive volume surges.
- **Inventory Costs:** Holding costs increase over time, incentivizing the AI to clear stock.

Section IV: Demand Forecasting

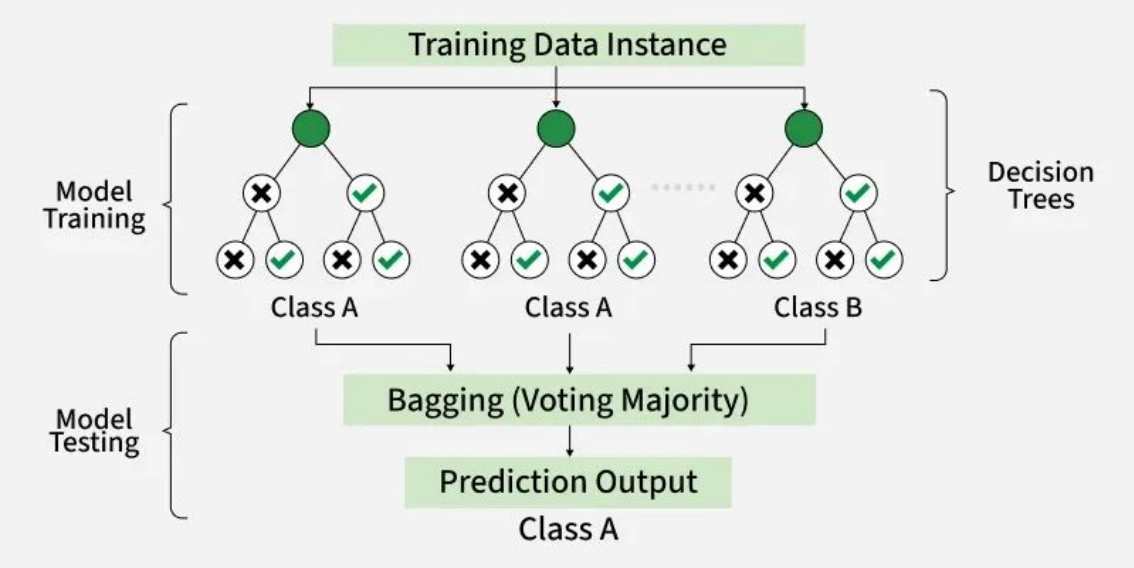
The "World Model": Predicting Consumer Response

Random Forest: The Market Oracle

Why Random Forest?

Random Forest was selected for its ability to handle non-linear relationships in retail data.

- **Ensemble Accuracy:** Aggregates predictions to reduce variance.
- **Feature Importance:** Clearly identifies if price or seasonality is the primary driver.
- **Robustness:** High resistance to overfitting synthetic noise.



Feature Engineering Strategy

Price Ratio

Relationship between our price and the competitor average.

Temporal Keys

Day of week, month, and seasonal binary flags.

Category Context

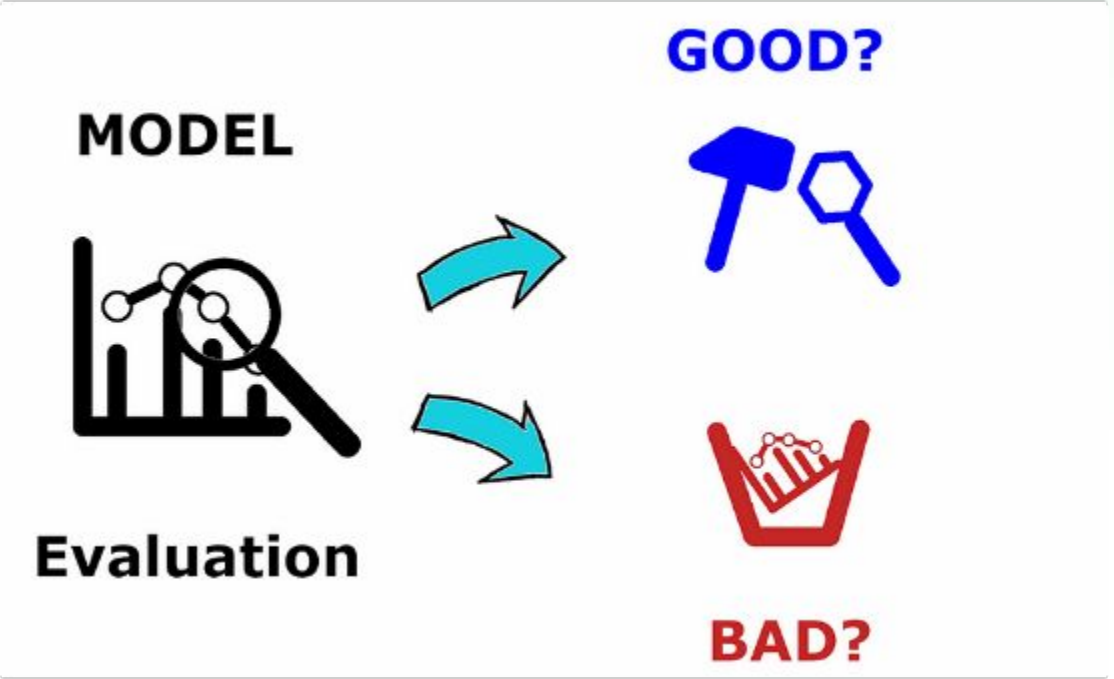
Embedding product categories to learn sector behaviors.

Forecaster Accuracy Metrics

0.94

R-Squared Accuracy

Highly reliable prediction of demand volume based on market state.



Section V: Price Optimization

Reinforcement Learning for Strategic Decision Making

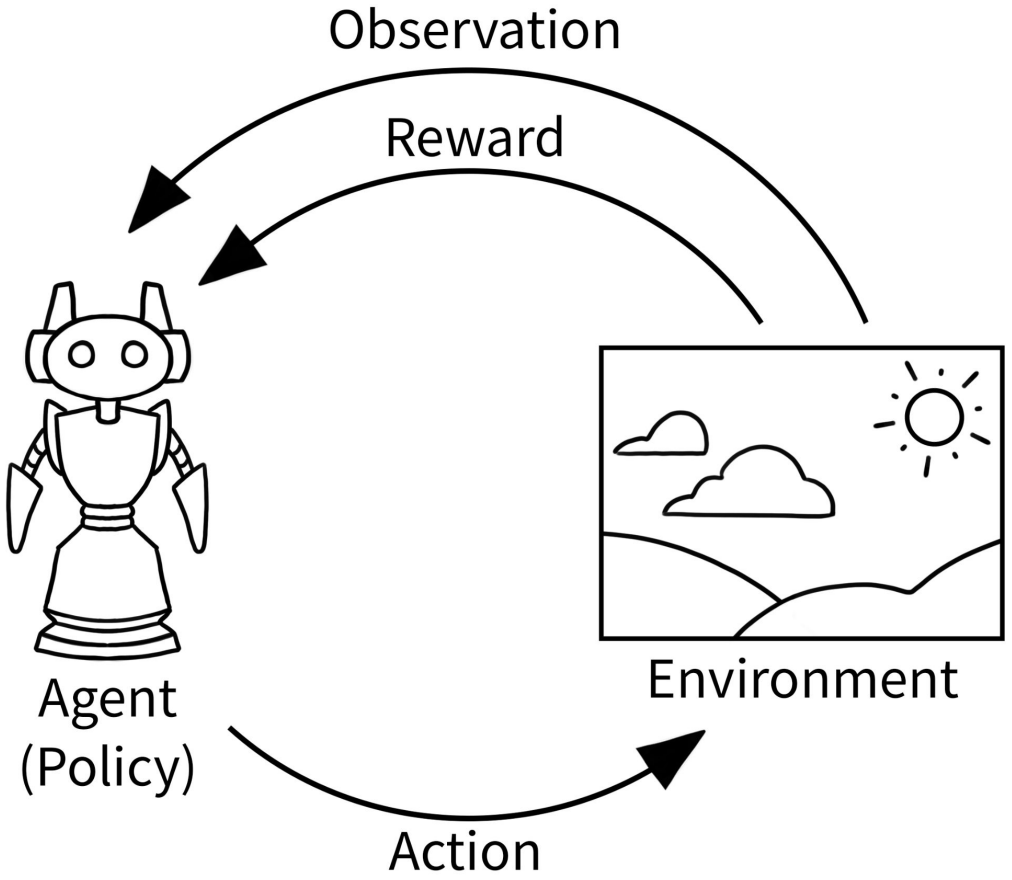
From Prediction to Policy

Sequential Optimization

Unlike standard regression, RL understands that a price change today affects demand tomorrow.

The agent learns to **position** itself against competitors to maximize long-term rewards.

Algorithm: Proximal Policy Optimization (PPO).



The Gymnasium Environment

State Space

Current Price, Competitor Price, Demand Forecast, and Product Cost.

Action Space

Continuous adjustment percentage of the current product price.

Transition

Stepping forward in time while the market state updates stochastically.

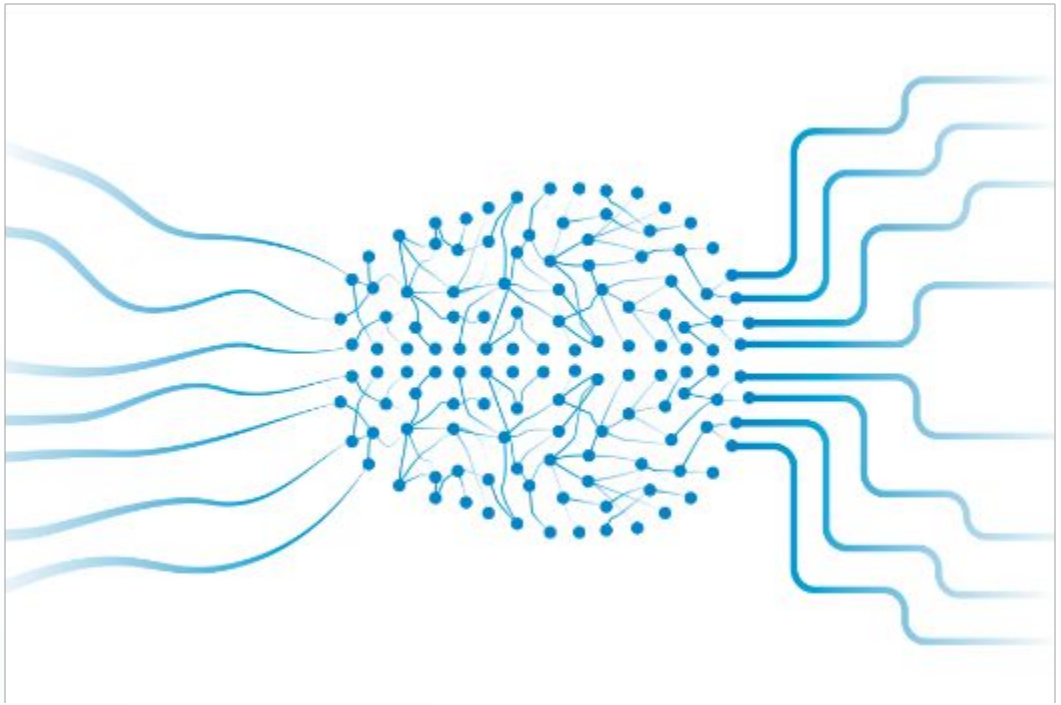
Reward Function Engineering

The agent's success is defined by its ability to balance Volume and Margin.

$$\text{Reward} = (P_{agent} - \text{Cost}_{SKU}) \times D_{predicted}$$

By maximizing this value, the agent naturally learns to lower prices for elastic items and premium-price essential goods.

Deep Policy Optimization (PPO)



Stable Reinforcement Learning

PPO was chosen for its reliability and "trust region" updates:

- **Continuous Action:** Fine-tuned price control.
- **Stability:** Prevents catastrophic collapses in policy during training.
- **Efficiency:** Rapid convergence in high-dimensional state spaces.

The "Model-in-the-Loop" Synergy

Strategic Integration

Our project doesn't just predict; it **simulates**.

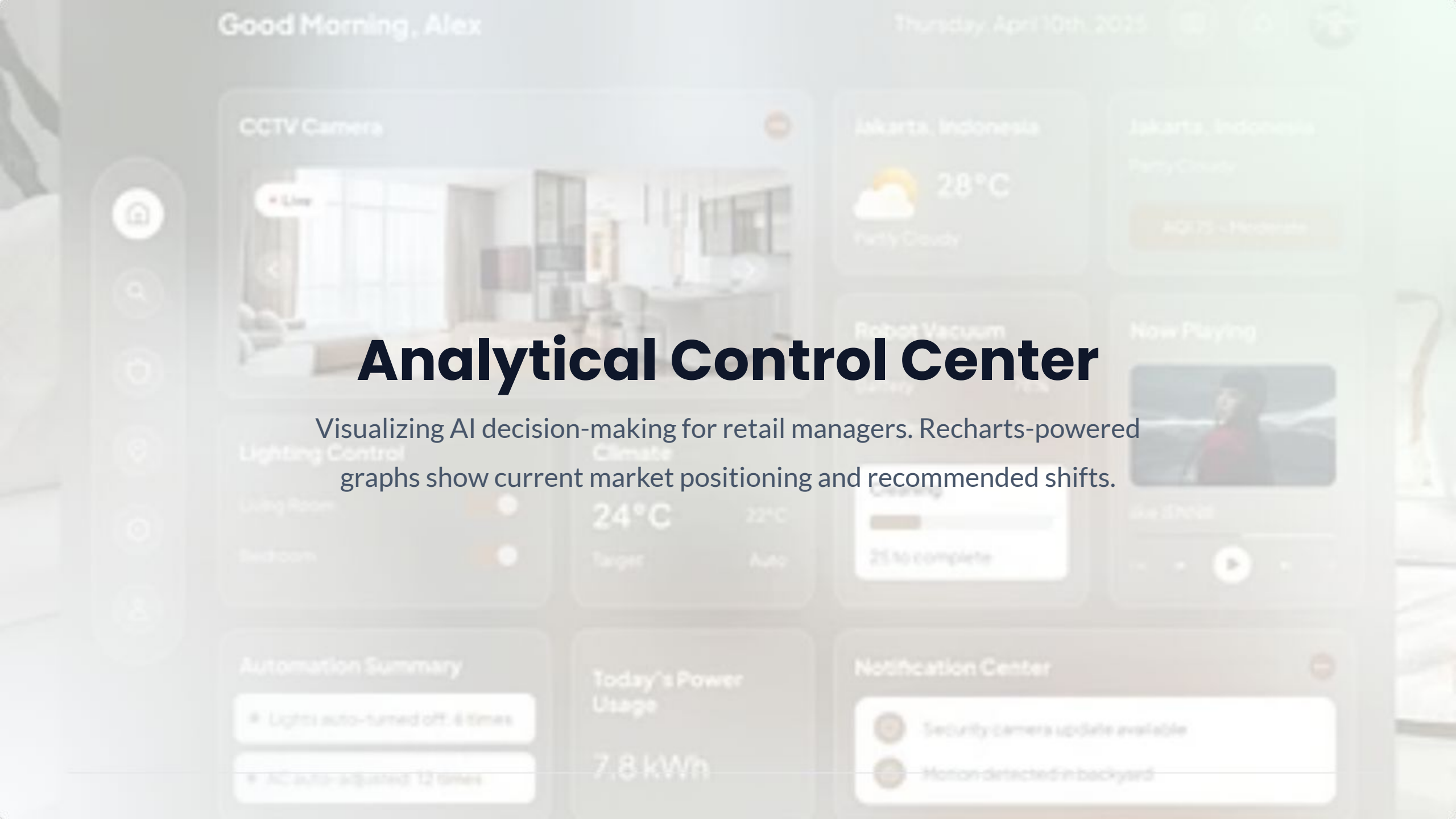
1. The Random Forest model creates a **"Mirror World"** of the market.

2. The RL Agent spends **millions of virtual days** in this Mirror World.

3. Once the Agent finds the winning strategy, it is **deployed** to the real-world dashboard.

Section VI: Software Implementation

A Professional Deployment Stack for Retail AI



Analytical Control Center

Visualizing AI decision-making for retail managers. Recharts-powered graphs show current market positioning and recommended shifts.

Section VII: Experimental Results

Quantifying the Impact of AI-Driven Pricing

Global Performance Lift



The PPO Agent consistently outperforms human-defined rules by finding non-obvious revenue opportunities.

Category-Level Impact

Product Category	Elasticity Score	Revenue Lift	Profit Margin Shift
Electronics	High (1.82)	+22.4%	+5.2%
Clothing	Medium (1.45)	+15.1%	+3.8%
Home & Garden	Low (1.12)	+11.8%	+2.1%
Beauty	Medium (1.56)	+18.3%	+4.5%

Agent Learning Stability



Policy Convergence: The PPO agent stabilizes after approximately 50,000 steps, learning the optimal pricing equilibrium for most product categories.

Final Conclusions

Autonomous pricing is no longer a luxury; it is a necessity for retail survival.

Scalability

The system easily handles new SKUs by learning their features from the initial database seed.

Adaptability

The "Model-in-the-Loop" architecture allows for retraining as market conditions shift.

Questions?

Thank you for your time.

Image Sources



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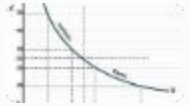
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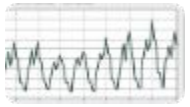
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